PCA Demo Quiz

~~* Indicates required question~~

1. **Q1 A:** *

Scenario: You have telemetry data from your FSAE car during the 22km endurance race. This includes 50 different sensor readings (e.g., all 4 wheel speeds, 4 suspension travel sensors, 4 tire temperatures, lateral force, steering angle, throttle position, motor torque, battery voltage, battery current, etc.) taken 100 times per second. This results in a massive spreadsheet with 50 columns and millions of rows.

Question: Your goal is to tell the driver which actions are the most important for a good lap time. Since there is no way to process that big of a file, you decide to use PCA to process the data. What would using PCA give you and how would you communicate that information to the driver?

(This question is designed to target an abstract and fundemental understanding of PCA and also the students ability to communicate it).

2. **Q1 B:** *

The driver is super happy to have this information but is really confused how you were able to extract such simple information from such a large dataset. Explain to the driver how you came to this conclusion and why your methods are trustworthy.

(This question is designed to target a students understanding of the PCA process and their ability to convery this complex topic simply. This shows a mastery of the process)

3. **Q2:** *

Scenario: You want to analyze how 100 different corporate brands post on social media. You collect all their posts for a year and measure 40 variables including: use of emojis, use of slang, color palette of images, response rate to comments, average post length, and "call to action" (e.g., "buy now," "learn more," "join us").

Question: You are a corporate advisor to a new start up looking to enter the market. How would you use PCA and your data set to help this company?

(*This question aims to get at practical uses of PCA. It forces the students to think outside of previously used contexts and apply math to the outside world. This will hopefully help them apply math more liberaly in the outside world.)*

4. **Q3:** *

Olin's admissions department wants to use PCA. They plan to combine 10 different data points from a student's application (grades, test scores, zip code, high school reputation, essays, SAT/ACT score) into a single "Principal Component 1 Score" to rank all applicants.

Explain why this system would unfairly favor some students. How would you fix this system to be more fair and representative?

(PCA inherently discards a lot of underrepresented data and can cover up underlying factors. This question probes students to find the unfairness of mathematical systems when applied in the wrong way.)

5. **Q4:** *

Your 16 year old brother just got his first smartphone, he is fascinated by its ability to recognize his face. You look really similar to him but the phone does not unlock for you. He knows that you are learning about the deep mysteries of facial recognition and asks you to explain it to him. Explain to him, in an engaging and clear way how his phone recognizes him and not you.

(While not exactly a PCA question, it generally evaluates PCA understanding in a practical application that is related to the course understanding of facial recognition.)

Mark only one oval.

- [ ] Option 1

6. **Q5:** *

Provide your funniest non-technical analogy for PCA.

(When students have fun when applying concepts they are more likely to remember them. Also, analogies are a great way to see if students have fundemental understanding of concepts.)

Mark only one oval.

- Option 1